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Research papers

An accurate and computationally efficient lithium-ion battery operation and degradation framework for planning and sizing studies in renewable energy systems

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ABSTRACT

Accurately modelling lithium-ion battery (LIB) degradation is essential for effective planning and sizing studies, yet the computational intensity of detailed day-by-day simulations poses practical challenges. This paper introduces two novel representative-day-based frameworks designed to balance computational efficiency with degradation accuracy. The first approach iteratively simulates the annual battery operation using representative days identified by clustering, while the second approach captures the initial rapid degradation through repeated cycles of these representative days, projecting long-term degradation efficiently. To verify the robustness and universality of the proposed methods, both conventional and proposed approaches have been tested using ten years of renewable generation profiles. Simulation results indicate that the first proposed approach achieves high accuracy with minimal deviation from detailed exhaustive simulations, completing multi-year assessments in under one minute compared to hours required by detailed day-by-day simulations. The second proposed approach further accelerates computations significantly, though with a moderate increase in error compared to the first proposed approach. Comprehensive sensitivity analyses demonstrate that the proposed methodologies provide a robust compromise between computational efficiency and accuracy, enabling practical and reliable LIB degradation assessment for planning and economic evaluations.

1. Introduction

1.1. Background and motivation

Lithium-ion batteries (LIBs) are widely deployed for stationary energy storage in grid applications due to their high power and energy density, and efficiency. The significant growth in battery energy storage systems (BESS) is evident globally; for instance, Australia's National Electricity Market (NEM) has seen an 86% year-on-year increase in its standalone BESS project pipeline, highlighting the rapid adoption and integration of these systems into modern power grids [1]. This growing adoption spans various applications, including utility-scale storage for energy arbitrage and grid support, ancillary services, commercial and residential peak demand shaving, and integration into microgrids characterised by high renewable energy penetration [2].

Every planning study involving LIBs necessitates careful consideration of battery degradation, as it directly influences the system's replacement costs and operational performance. LIBs typically exhibit relatively short lifetimes, significantly influenced by their operational patterns. Accurate estimation of battery degradation is crucial to ensure long-term reliability and economic viability of these storage systems, given the complex interplay between battery usage patterns and underlying degradation mechanisms. Underestimation of battery degradation can substantially affect the economic feasibility and operational efficiency of grid-scale storage deployments [3].

This paper does not propose a new battery degradation model, but instead presents frameworks designed to estimate battery degradation using any existing degradation model for planning and sizing studies. These frameworks are particularly tailored for net present cost (NPC) analysis studies involving LIBs. Typically, modelling LIB degradation in grid applications involves two contrasting approaches: (i) full-resolution models, which simulate degradation at daily or hourly intervals throughout the battery's lifetime, and (ii) computationally efficient representative day approaches, which reduce complexity by selecting a subset of days hypothesised to capture the statistical characteristics of the complete dataset [4,5]. While full-resolution simulations provide the highest accuracy, their computational demands can be prohibitive, especially when integrated into large-scale optimisation

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studies. Representative day methodologies offer substantial computational savings, but they can introduce inaccuracies in capturing extreme operating conditions, deep cycling events, and seasonal variations [6].

1.2. Literature review

The existing literature extensively discusses various methodologies for analysing battery degradation in studies focused on storage. For instance, Sagaria et al. [7] employed a detailed simulation in MATLAB/Simulink over a 10-year period to analyse battery degradation impacts due to vehicle-to-grid (V2G) operation. It simulates the first year, then runs iteratively year by year, updating cycle ageing and capacity fade until the battery reaches its End-of-Life (EoL) (e.g., 80% capacity remaining). Similarly, Shabani et al. [8] conducted battery degradation analysis for residential battery application by simulating hourly battery operation over a full year and calculating the capacity fade, repeating the process annually until the EoL of the battery. They examined 32 distinct battery operating control strategies characterised by varying allowable State-of-Charge (SoC) windows, without employing representative days. Instead, the authors performed full-year simulations iteratively to explicitly evaluate the impacts of calendar and cyclic degradation, subsequently assessing the lifetime and economic outcomes for residential battery applications.

In other studies, Hollinger et al. [9] analysed battery degradation in a residential BESS by performing simulations using a full-year time series profile at a resolution of one second. Rather than simulating battery operation directly until the EoL, they assessed battery lifetime and degradation impacts based on this detailed, single-year simulation profile without employing representative days. The battery simulations in this research were initialised by assuming full battery capacity, reflecting operational behaviour during the initial year of the battery lifecycle under ideal health conditions. In the same vein, Hesse et al. [10] performed battery degradation simulations using a full-year time series profile with 15-minute intervals for residential PV-battery energy storage systems. Their simulations were conducted over one year and, rather than extending the analysis directly to the battery EoL, the battery life and degradation were estimated by extrapolating from the observed cyclic and calendar ageing data over one year. Similarly, this study considered the battery's initial state by assuming that it was at full capacity at the beginning of the simulation.

Since battery degradation occurs gradually, analysing degradation on a day-to-day basis can provide more detailed insights compared to yearly analyses. For instance, Narayan et al. [11] developed a practical battery lifetime estimation model specifically for solar home systems, conducting battery degradation analysis through day-to-day simulations spanning one year, without extending the simulation directly to the battery's EoL. They employed two approaches, a coarse average method and a more detailed zero-crossing (ZC) micro-cycle method, both applied over the one-year simulation period. Rather than using representative days or explicitly simulating until battery EoL, they approximated battery lifetimes by extrapolating from the single-year simulation results. In this study, battery degradation was modelled based on the initial battery condition, assuming full battery capacity (initial battery health) at the simulation's outset. Similarly, Aaslid et al. [12] employed a multi-stage stochastic programming approach combined with rolling-horizon optimisation to simulate battery operation on a daily basis for a full year. They explicitly incorporated both Depth-of-Discharge (DoD) and SoC impacts on degradation into their analysis, but their simulations stopped after one year and did not extend directly until the battery's EoL. Similar to the previous study, this study also conducted simulations for a single year of battery operation, explicitly considering the battery degradation model based on its initial full battery capacity. Another study conducted by Marańda [13] analysed battery degradation through day-to-day simulations using real data over a full one-year period. In this study, battery lifetime or EoL was not directly simulated; instead, annual capacity degradation was estimated based on cycle counts and DoD derived from the detailed one-year dataset. Similarly, this study specifically considered a 100% battery capacity for their simulation studies. In a similar vein, Fortenbacher et al. [14] analysed battery degradation comprehensively for optimal sizing and placement of battery storage in distribution grids by employing an iterative Receding Horizon Control (RHC) strategy within a Model Predictive Control (MPC) framework. They simulated battery operation sequentially over shorter control horizons (6 h, 12 h, daily, weekly, and monthly), rather than performing a single full-year simulation, dynamically incorporating load and PV generation forecasts to enhance decision-making accuracy. Battery EoL was not directly simulated; instead, it was predicted by extrapolating degradation data obtained from the one-year simulation, considering both DoD and SoC effects. Likewise, this study specifically considered the battery's initial full capacity, thus reflecting the battery's initial health condition at the start of the simulation.

In contrast, only a limited number of studies in the literature have accurately conducted battery degradation analysis by performing day-to-day simulations until the EoL. However, these studies incurred significantly high computational costs. As demonstrated by Casals et al. [15], they used detailed day-to-day simulations for various stationary applications to evaluate battery degradation until the EoL was reached. They specifically analysed scenarios including electric vehicle (EV) fast charging, self-consumption, frequency regulation services, and transmission deferral, applying an equivalent electric battery ageing model validated by accelerated laboratory ageing tests. Their simulations covered extensive periods (up to 20 years) to precisely estimate the battery's remaining useful life under realistic operating conditions. Likewise, Sowe et al. [16] analysed battery degradation by conducting hourly simulations iteratively over a 20-year operational horizon, rather than simulating a full year in one go or employing representative days. They used a semi-empirical battery degradation model that incorporated detailed calendar and cycle ageing mechanisms influenced by temperature and SoC. Battery EoL was explicitly predicted through this iterative long-term simulation approach, identifying how different operational strategies, such as SoC-based derating, significantly influenced battery lifetime extension.

Although analysing battery degradation over extended periods provides detailed insight, the high computational complexity makes it impractical for many applications, such as optimal sizing and planning of battery storage systems. Consequently, many recent studies have used the representative days approach to accelerate battery degradation analysis while improving computational efficiency. However, the accuracy of this approach remains a subject of debate. In one study, Salek et al. [17] developed a mathematical model for second-life battery packs and performed battery degradation simulations using representative days of each season (spring, summer, autumn, winter) to approximate the full annual operational profile. Specifically, they selected four representative days, one for each season, to simulate battery degradation over a one-year period, effectively capturing seasonal variations in charging and discharging conditions. The annual degradation results obtained from these representative-day simulations were then extrapolated to estimate the battery's EoL. In contrast to previous literature, this study performed simulations using one year of operational data, explicitly considering the terminal battery health condition (the last year of battery operation) rather than the initial full battery capacity. This conservative approach is crucial, as it reflects the realistic starting point of aged batteries and ensures degradation assessments account for worst-case scenarios—an essential consideration in robust planning and lifecycle cost evaluation. In another study, Uddin et al. [18] performed battery degradation simulations for the V2G application of EVs in smart grids using real-world usage cycles, each representing one week. Rather than simulating the battery directly over its entire lifetime, they extrapolated annual degradation from these representative weekly cycles to predict longer-term effects and potential EoL conditions. In contrast, in this research, the battery is

Table 1
Summary table of all relevant literature.

Ref.	Single year		Representative days	Simulated until EoL	Day-to-day operation	Full year continuously
	LIB full capacity (Single year)	LIB terminal condition (Last year)				
[7], [8]	✗	✗	✗	✓	✗	✓
[9], [10]	✓	✗	✗	✗	✗	✓
[11], [12], [13], [14]	✓	✗	✗	✗	✓	✗
[15], [16]	✗	✗	✗	✓	✓	✗
[17]	✗	✓	✓	✗	✓	✗
[18], [20]	✓	✗	✓	✗	✓	✗
Our study	✓	✓	✓	✓	✓	✓

modelled starting from its initial capacity (full battery health) rather than adopting a terminal (last-year) capacity perspective. Likewise, Schneider et al. [19] used representative days (specifically, 10 typical days) to approximate a full-year battery operation profile for peak demand shaving and energy arbitrage rather than performing continuous day-to-day simulations. They integrated the Rainflow cycle counting algorithm into a multi-objective optimisation framework to accurately capture cyclic battery degradation, and subsequently extrapolated annual degradation estimates obtained from these representative simulations to predict the battery's EoL. Moreover, this study modelled the battery from its initial capacity. Furthermore, Schram et al. [20] analysed battery degradation for optimal sizing of residential PV and storage systems by performing simulations using measured data for 295 representative days, evenly distributed throughout the year, at a 10-second interval. They did not simulate until the battery reached EoL, instead assessing battery economic performance based on a one-year simulation. While such short-term evaluations offer computational advantages, they may overlook the long-term nonlinear nature of battery degradation—particularly the fact that degradation tends to accelerate exponentially as the battery approaches EoL. Accurately capturing this behaviour is essential for reliable lifetime and cost assessments, especially in planning studies where underestimating late-stage degradation could lead to suboptimal investment and operational decisions. Their study used two battery control strategies, heuristics and perfect foresight, to evaluate peak shaving potential but relied on economic indicators rather than direct lifetime predictions. Similar to other studies, the present work began with the initial (full) battery capacity. Although these methods are computationally efficient, none of the aforementioned studies validated the accuracy of the battery degradation values obtained solely through representative-day simulations. Table 1 summarises all relevant literature in comparison to our study.

1.3. Research gaps and contributions

The literature review highlights that representative day techniques are widely adopted for modelling various energy systems, primarily because of their computational efficiency in handling long-term, high-resolution simulations. Despite their widespread use, existing research has not systematically quantified the impact of these representative day methods on the accuracy of battery degradation predictions. Accurate degradation modelling is crucial for planning and operational analysis of battery energy storage systems, as inaccuracies, whether through under- or overestimation, can compromise battery life estimation, leading to suboptimal system designs and operational strategy leading to incorrect estimates of costs and benefits. This could ultimately affect the economic viability and operational efficiency of grid-scale storage. Many studies in the literature have thus far simulated only a single year of battery operation, typically assuming the battery starts at its initial full capacity, but a more practical approach for single-year simulations would be to assume the battery is already at its terminal health (i.e., last year of operation). This pragmatic assumption guarantees sufficient battery capacity to meet its needs until EoL. It also provides a more precise daily estimation of battery performance as the capacity gradually degrades, rather than assuming a constant capacity throughout the yearlong simulation.

Simulating only one year of battery operation without complex control strategies typically incurs manageable computational time. However, when more advanced operational strategies — such as receding-horizon optimisation — are implemented, even a single-year analysis can become computationally intensive. Furthermore, in hyper-realistic operational studies and techno-economic analyses that require iterating simulations over multiple years until EoL, the computational burden grows dramatically. In these multi-year contexts, representative day approaches become essential. By reducing the number of days simulated at full resolution, they allow detailed degradation patterns and economic implications to be captured without the prohibitive cost of running high-resolution day-by-day simulations over the entire battery lifetime. Hence, there is a clear lack of robust frameworks that:

- Balance computational efficiency and accurate estimation of battery degradation in planning studies
- Generalise well across different storage applications
- Provide a flexible framework for effectively estimating battery degradation at much lower computational time, without relying on a new or proprietary degradation models, i.e., compatible with any existing model

To address these research gaps, this paper explores different approaches to propose a computationally efficient yet accurate framework for battery degradation estimation framework using representative days selected by clustering techniques. It has the following key contributions:

- Development of two novel approaches to efficiently model long-term LIB degradation without incurring prohibitively long simulation times. Both approaches leverage representative days, but use different strategies to use representative days over multiple years. By capturing key degradation phenomena in a fraction of the time required by detailed day-by-day simulations until the EoL, they offer a practical way to investigate multiyear battery behaviour. Below are the details of each approach:
  - **Proposed Approach I:** In this approach, a small number of representative days is initially selected using clustering techniques. The battery operation for each representative day is then individually optimised through simulation, and the results from these simulations are subsequently replicated according to each day's assigned weight to construct an accurate full-year operational profile. Based on this reconstructed annual profile, the battery's yearly degradation is calculated and the battery's energy capacity is updated to reflect the accumulated capacity loss. This process is repeated iteratively for subsequent years until the battery reaches its predefined EoL capacity threshold.

– **Proposed Approach II:** In the second proposed approach, a set of representative days is identified similar to Proposed-I by clustering techniques and the battery operation for each representative day is again individually optimised through simulation. Following the first year, rather than presuming that the battery operates at full capacity at the beginning of each subsequent year, the battery capacity value is instead recalculated annually, with updates applied at the end of each year. In this case, the battery operation of each representative day is replicated for a fixed number of consecutive days to effectively capture the linear portion of battery degradation. The reason is that batteries typically have higher initial degradation rates that subsequently stabilise with extended use. The linear degradation value of this extended period for each representative day is then multiplied by its assigned weight. Summing these weighted degradation values provides the total annual degradation, which is then used to determine the battery EoL by extrapolation.

It is not possible to analytically determine which approach is universally superior, as each may perform optimally in some cases and applications, but not in others. In the cases where either method is suitably applied, it should:

- Significantly reduce simulation time while retaining high degradation accuracy
- Accurately analyses battery degradation over multiple years until EoL
- Showcase minimal deviation in battery replacement schedules and net present cost

1.4. Paper organisation

The rest of the paper is organised as follows: Section 2 introduces the methodology, presenting details on the representative day techniques based on clustering, the frameworks for implementing the conventional and proposed methodologies, the LIB operation and degradation model, the economic cost modelling approach, simulation tools and environment. Section 3 discuss about the simulation setup and input parameters about the case study. Section 4 presents and analyses the results, comparing battery degradation, computational efficiency, replacement schedules, and net present cost (NPC) in different approaches. Finally, Section 5 concludes the paper, summarising key findings and outlining directions for future research.

2. Methodology

2.1. Representative days approach and clustering methods

A multiyear, day-by-day simulation at high resolution until the battery reaches EoL, hereafter referred to as the hyperreal approach, is widely considered the most accurate method for capturing LIB degradation under realistic operational conditions. However, beyond being prohibitively time-consuming for large-scale or long-duration storage systems over multiple years, this exhaustive framework also cannot be readily integrated into mathematical optimisation studies due to the sheer size of the resulting problem and the nonlinear nature of degradation models. To address this issue, it is crucial to develop a method that significantly reduces the time horizon but preserves the finer operational details that dictate battery ageing. Representative day strategies offer a promising route to achieve these efficiency gains; however, if not carefully designed and validated against the 'Base' case day-to-day full resolution simulation, they risk overlooking critical yet infrequent operating conditions such as deep cycling, extreme temperatures, and seasonal variations. This could lead to substantial inaccuracies in estimating battery degradation, which ultimately affects decisions on battery planning and sizing, scheduling, and net present cost estimation. Consequently, there is a pressing need for a more robust framework that estimates LIB degradation very close to the 'Base' case degradation value while reducing the computational burden significantly, ensuring that degradation estimates remain precise and practical for real-world applications.

To significantly reduce computational burden while retaining the essential operational characteristics that drive battery degradation, we cluster the original annual dataset of 365 daily profiles (five-minute timestep, total of 105120 timesteps) using the K-means algorithm. Among the various clustering techniques (e.g., K-means, K-medians, K-centres, K-medoids, seasonal-based, and monthly-based methods), the K-means method has consistently been shown to provide superior accuracy in capturing original load or net demand patterns, as evidenced by Schütz et al. [21], Novo et al. [22], and Orgaz et al. [23]. These studies reported lower sum of squared errors, reliable results even with fewer representative days, and significant reductions in computational complexity, all indicating that the K-means method is robust and computationally efficient. This method partitions the full set of daily profiles into a predefined number of clusters, each sharing similar net demand patterns. From each cluster, a 'typical' daily profile, often the cluster centroid or a close approximation, is selected as the representative day. Each representative day is then weighted by the number of actual days it encapsulates, reflecting the frequency with which its characteristic pattern occurs throughout the year. By substituting the full year dataset with a handful of representative days, accompanied by their respective weights, the simulation horizon is compressed dramatically. Nevertheless, key daily variations and seasonal cycles remain captured, ensuring that the resulting degradation estimates accurately mirror those that would be obtained from a more exhaustive day-by-day simulation approach.

2.2. Framework of the proposed methodology

Fig. 1 illustrates the overall structure of our proposed methodology, beginning with the input parameters, namely load profiles, energy resource data, and essential battery specifications such as capacity and C-rate. These inputs are fed into the "Battery Operation Framework", within which the battery is operated according to one of several approaches: Base Case, four Conventional (I–IV) approaches, or the two Proposed (I–II) approaches, each briefly summarised in Section 2.3 and illustrated in Fig. 2. Once the battery operation is simulated and the corresponding SoC and temperature profiles are obtained, these outputs can be fed into a battery degradation analysis tool — such as BLAST (Battery Lifetime Analysis and Simulation Tool) or any other suitable model — to quantify ageing over time. For instance, BLAST (publicly available as a Python library [24]), developed by the National Renewable Energy Laboratory (NREL) in the USA, captures both calendar and cycle ageing. However, any validated degradation model that accepts SoC and temperature inputs may be used in place of BLAST to estimate the extent of battery degradation.

In the operation framework, each approach differs in computational requirements when simulating battery operation and degradation. The Base Case delivers the most precise estimation, but also has the longest run time. In contrast, both the conventional and proposed methods are significantly more time-efficient. Our objective is to compare their accuracy against the base case to investigate the trade-off between accuracy and computation time.

2.3. Brief description of the approaches

This section presents a brief description of the Base Case, Conventional I, II, III, IV and Proposed I, II approaches, which have been shown in Fig. 2.

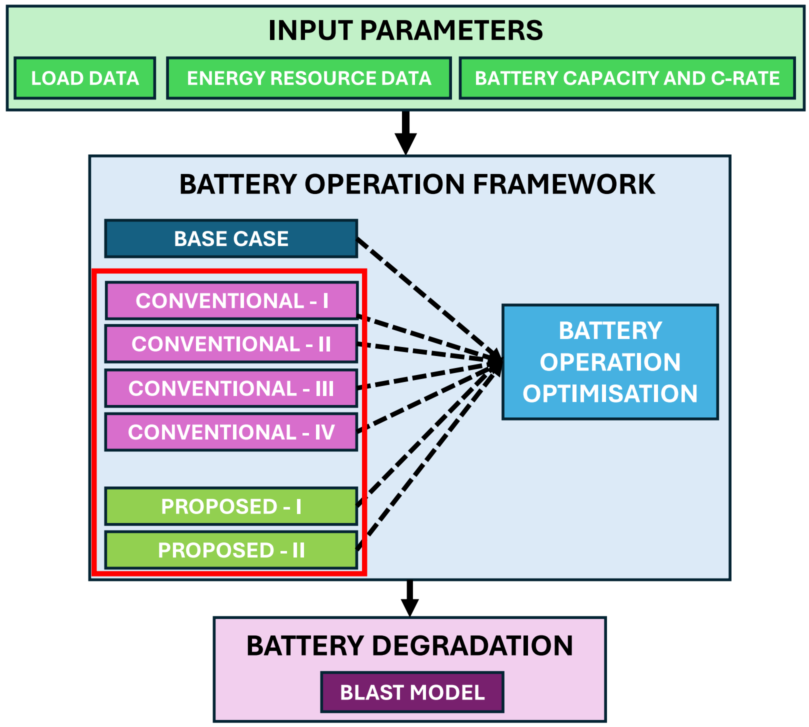


Fig. 1. High-level flowchart of the proposed methodology.

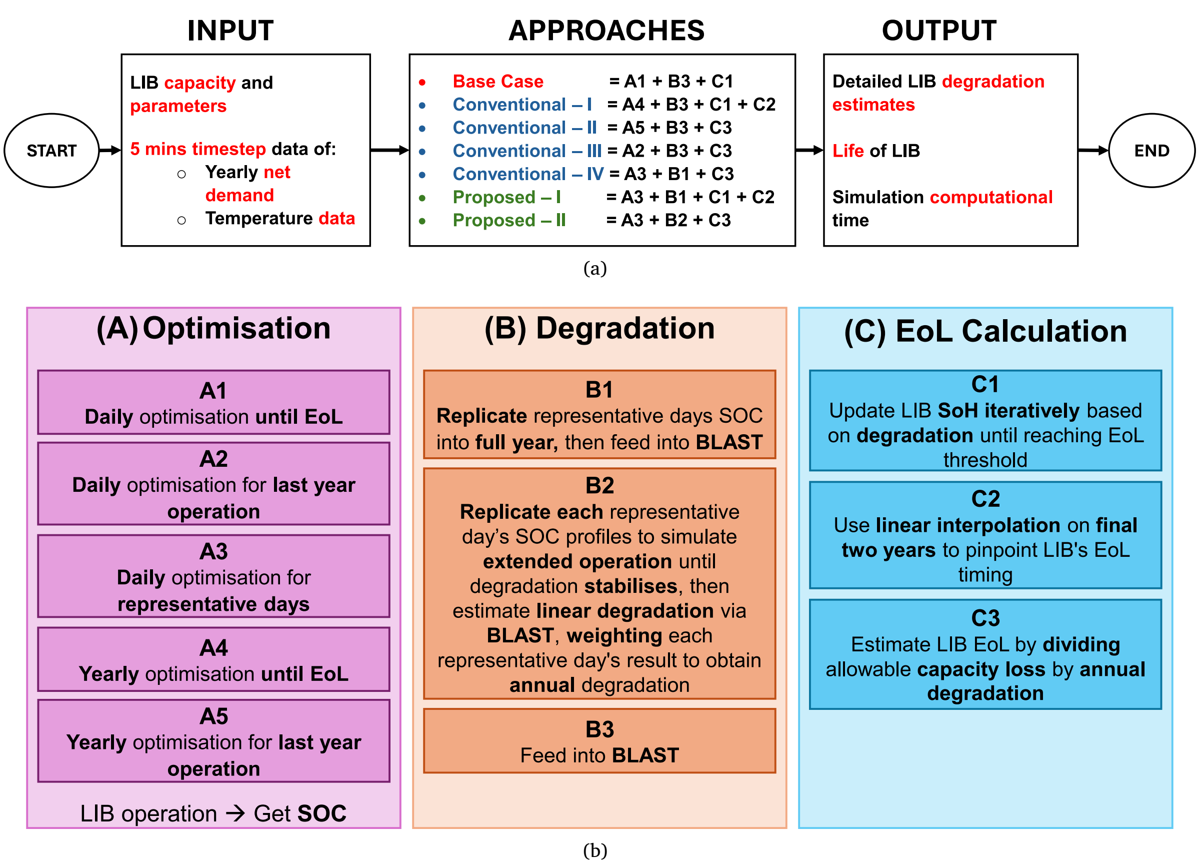


Fig. 2. Overview and explanation of the different approaches' framework; (a) shows the overall structure, while (b) provides a brief explanation of the sub-frameworks.

Three basic principles are common to all methods: (1) net demand data with five-minute resolution, (2) a LIB EoL threshold at 80% of the initial capacity, and (3) systematic recording of the SoC profile of each horizon and updated maximum energy capacity for the next horizon.

- **Base Case [A1+B3+C1]** [15,16]: The battery is simulated *day-by-day*. Each day starts with the capacity carried over from the previous day, and the cycle repeats for all 365 days. When the calendar year ends, the model loops back to Day 1 and continues through successive years *day-by-day* of the same profile until the relative capacity drops to 80%. In the Base Case, this approach therefore solves $365 \times N + n$ optimisation problems (where N is the number of full years to EoL and n is the remaining days in the last year), and because of this, the Base Case takes the longest computational time. Dufo-López et al. [25] and DiOrio et al. [26] demonstrate that repeating a single meteorological year can be effective when only one year of data exists, however many studies — including ours — have access to multiple years of historical profiles. In such cases, experts often select the "worst-case" year (or repeat a representative worst year) because the specific sequence and ordering of annual conditions can dramatically accelerate degradation and may not recur identically in the future. By focusing on this conservative scenario, the repeated annual profile still yields a precise capacity-fade trajectory and an exact day count to EoL, while avoiding reliance on a historical sequence that may not reflect real-world operational patterns.
- **Conventional-I [A4+B3+C1+C2]** [7,8]: Similar to the Base Case, this approach runs the battery all the way to EoL with capacity updates after each horizon; however, the horizon is one full year rather than one day. Capacity is held constant for 365 days, then reduced once at the end of each year, and the process repeats annually. In Conventional-I, this approach therefore solves $N + 1$ optimisation problems (one per year plus one for the remaining days), and because this is much fewer than the Base Case's $365 \times N + n$ optimisations, it takes significantly less time. Specifically, the optimisation problem is solved once for the entire year — rather than solving 365 daily optimisations as in the Base Case — so runtime is cut substantially. The exact *year count* to EoL is then obtained by linearly interpolating between the last two simulated years.
- **Conventional-II [A5+B3+C3]** [9,10]: Building on Conventional-I, this method still uses a single full-year horizon but does not iterate until EoL. It treats that lone year as the final operational year, solves one optimisation problem, records the resulting annual degradation rate, and then estimates lifetime by dividing the allowable 20% fade by that rate. Because only a single full year optimisation problem is required, computation is very fast—though at the expense of an approximate lifetime projection.
- **Conventional-III [A2+B3+C3]** [11–14]: This variant merges features of the Base Case and Conventional-II. It operates every day separately (as in the Base Case) but without updating the capacity each day and limits the simulation to a single calendar final operational year and infers EoL from the aggregated annual fade, as in Conventional-II. Hence, this approach solves 365 daily optimisation problems, thus balances daily operational detail with reduced runtime.
- **Conventional-IV [A3+B1+C3]** [17–20]: Conventional-IV follows the same one-year-only, analytical-lifetime logic of Conventional-II, but replaces the full 365-day record with a small set of weighted representative days obtained through K-means clustering, or any other clustering techniques. Those weighted days are concatenated into a synthetic "year", from which the annual degradation rate is calculated and then scaled to lifetime. Clustering further lowers computational effort since it solves daily optimisation for only representative days, however it introduces a sampling error that must be acknowledged.
- **Proposed-I [A3+B1+C1+C2]:** Proposed-I follows multiyear simulation until EoL. The same weighted representative days form a synthetic year, then that synthetic year is simulated year-by-year until EoL, with capacity updated after each year and linear interpolation between the final two years to pinpoint the exact lifetime. The approach preserves the runtime savings of the representative day approach while capturing multi-year capacity evolution. The detailed implementation steps of this approach are explained and outlined in the later part of the paper in Algorithm 1.
- **Proposed-II [A3+B2+C3]:** Sharing the Proposed-I representative day foundation, this approach first "settles" each cluster by cycling every representative day repeatedly (e.g. hundred days) to let the higher initial fade stabilise. This is because LIB has very high degradation at initial days and later stabilises, as illustrated in Fig. 3. The final day's degradation is taken as the characteristic value for that day type; multiplying by the day's weight yields the annual degradation, and lifetime is again estimated analytically. Since it solves daily optimisations only for a handful number of representative days, as a result, Proposed-II retains minimal computational cost while accounting for the early-life non-linear capacity degradation that is typical of LIB cells. The detailed implementation steps of this approach are explained and outlined in the later part of the paper in Algorithm 2.

2.4. Battery operation framework

It should be noted that the battery operation module described here represents a general optimisation framework and could, in principle, be replaced by any other suitable formulation depending on the application context. In this study, we adopt the specific formulation developed for our mining microgrid problem as a case study. The objective, constraints, and decision variables presented below are therefore tailored to the particular characteristics of the mining use case—even though the overall structure can be generalised to other battery-operation problems in different settings.

2.4.1. Objective function

The primary goal of battery operation is to minimise the total energy not served and renewable energy curtailment. However, reducing the amount of energy not served is typically more critical than limiting renewable energy curtailment. Hence, the objective function has been carefully structured to prioritise the minimisation of energy not served over the curtailment of renewable generation, ensuring the most effective utilisation of available resources and enhanced reliability of the microgrid operation.

$$\min \left[\sum_{t=1}^{T} \left(\beta_{PF}(t) \cdot P_{ns}(t) + \alpha_{PF}(t) \cdot P_{curt}(t) \right) \times \Delta T \right] \tag{1}$$

The objective function in (1) aims to minimise the sum of power not served (P_{ns}), and the renewable curtailment (P_{curt}), each weighted by a penalty factor, β_{PF} and α_{PF}, respectively. The penalty factor β_{PF} is assigned a much higher value than α_{PF} to ensure that minimising P_{ns} is given the highest priority. The penalty factors β_{PF} and α_{PF} gradually decrease from the initial time steps to later/final time steps, which encourages the battery to discharge immediately when power is not served, avoiding situations in which the battery remains unused despite having sufficient SoC. Meanwhile, α_{PF} penalises power curtailment in a similar way.

2.4.2. Constraints

Energy balance: The energy balance constraints (2)–(5) ensure that the microgrid's power generation and consumption remain balanced in each time step t.

$$P_{Li}^{\text{dis}}(t) + P_{ns}(t) = P_{net}(t) + P_{Li}^{\text{ch}}(t) + P_{curt}(t), \quad \forall t \tag{2}$$

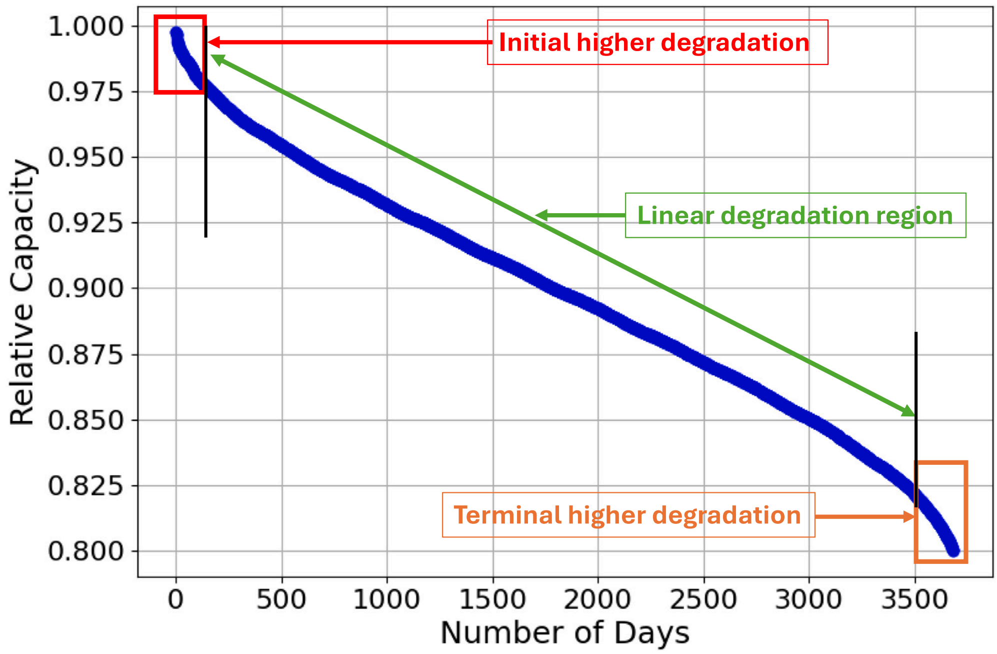


Fig. 3. LIB degradation pattern throughout days until EoL.

$$P_{net}(t) = P_{load}(t) - P_{solar}(t) - P_{wind}(t), \quad \forall t \tag{3}$$

$$0 \le P_{curt}(t) \le P_{solar}(t) + P_{wind}(t), \quad \forall t \tag{4}$$

$$0 \le P_{ns}(t) \le P_{load}(t), \quad \forall t \tag{5}$$

Constraint (2) represents the balance between supply and net demand ($P_{net}(t)$). At any given time t, the load must be met by the available solar and wind generation, supplemented by either discharging the battery ($P^{dis}_{Li}(t)$), or charging the battery ($P^{ch}_{Li}(t)$) and allowing some renewable power to be curtailed. If the combination of renewables and battery discharge cannot fully satisfy the load, the unmet portion is considered $P_{ns}(t)$.

The constraint (3) defines $P_{net}(t)$ as the difference between the load demand (P_{load}) and the renewable energy available from solar ($P_{solar}(t)$) and wind ($P_{wind}(t)$). This net demand can vary, being positive when the load exceeds the renewable generation or negative when renewable generation is surplus.

Constraints (4) and (5) impose practical limits on renewable curtailment and power not served, respectively. Specifically, renewable power curtailment cannot exceed the total renewable generation available at that timestep, ensuring no artificial surplus. Likewise, the power not served cannot exceed the actual load in any timestep, ensuring that the simulation reflects realistic operating conditions.

Battery operation: The battery operation constraints (6)–(9) govern the LIB's charging and discharging behaviour, as well as the State of Energy (SoE) throughout the simulation period.

$$SoE_{Li}(t) = SoE_{Li}(t-1) + \Delta T \cdot \left(\eta^{ch}_{Li} \cdot P^{ch}_{Li}(t) - \frac{P^{dis}_{Li}(t)}{\eta^{dis}_{Li}} \right), \quad \forall t \tag{6}$$

$$0 \le P^{ch}_{Li}(t) \le P^{max}_{Li} \cdot u_{Li}(t), \quad \forall t \tag{7}$$

$$0 \le \frac{P^{dis}_{Li}(t)}{\eta_{Inv}} \le P^{max}_{Li} \cdot (1 - u_{Li}(t)), \quad \forall t \tag{8}$$

$$u_{Li}(t) \in \{0, 1\}, \quad \forall t \tag{9}$$

*Constraints for running **until EoL** scenario:*

$$\alpha^{min}_{SoE} \cdot E^{max}_{Li} \le SoE_{Li}(t) \le \alpha^{max}_{SoE} \cdot E^{max}_{Li}, \quad \forall t \tag{10}$$

*Constraints for running **single year** scenario:*

$$E_{Li}(t) = E^{max}_{Li} \cdot SoH^{E-1}_{Li} - \frac{(E^{max}_{Li} \cdot SoH^{E-1}_{Li} - E^{max}_{Li} \cdot SoH^{E}_{Li})}{T} \cdot t \tag{11}$$

$$\alpha^{min}_{SoE} \cdot E_{Li}(t) \le SoE_{Li}(t) \le \alpha^{max}_{SoE} \cdot E_{Li}(t), \quad \forall t \tag{12}$$

Additional constraint for simulating day to day operation:

$$SoE_{Li}(0) = SoE_{Li}(L_d) \tag{13}$$

Eq. (6) tracks the battery SoE at each time t. Specifically, it updates the battery current SoE ($SoE_{Li}(t)$) based on its previous state ($SoE_{Li}(t-1)$), accounting for the energy gained through $P^{ch}_{Li}(t)$ and the energy delivered through $P^{dis}_{Li}(t)$. The charging and discharging power values are adjusted by their respective efficiencies (η^{ch}_{Li} and η^{dis}_{Li}) to accurately represent real-world operational losses.

The constraints (7) and (8) limit the battery's charging and discharging rates. The battery cannot charge or discharge beyond its maximum power capacity (P^{max}_{Li}), η_{Inv} considers the efficiency of the inverter, and the binary variable $u_{Li}(t)$ in Eq. (9) ensures that the battery is charging or discharging, but not both simultaneously.

In the running until EoL scenario, Eq. (10) assumes that the LIB operates from its maximum rated energy capacity E^{max}_{Li}. In this scenario, the LIB's $SoE_{Li}(t)$ is consistently bounded within predefined operational limits, defined by α^{min}_{SoE} and α^{max}_{SoE}. Since the simulation runs iteratively across multiple years until the battery reaches its EoL, it starts from full capacity, and the battery's maximum available capacity is progressively updated each subsequent day or year based on accumulated degradation.

In contrast, the single-year scenario uses a linearly decreasing energy capacity as defined in (11). Here, the simulation explicitly considers only a single operational year, conservatively assumed as the battery's final year of life. SoH^{E-1}_{Li}, SoH^{E}_{Li} are the State of Health (SoH) of LIB at the beginning of the last year and EoL, respectively. Thus, the battery begins to operate at reduced capacity, representative of its terminal health. Eq. (12) subsequently maintains the battery energy state within safe operational limits, determined by the dynamically changing maximum capacity $E_{Li}(t)$.

Eq. (13) introduces an additional constraint specifically for day-to-day battery operation, including scenarios that use representative days, ensuring that each day's operation is handled independently, unlike the simulation which runs an entire year in one continuous run. The purpose of this constraint is to ensure that the battery SoE in the first $SoE_{Li}(0)$ and last $SoE_{Li}(L_d)$ time steps of each day remains the same.

LIB Number of Cycles: The constraint in Eq. (14) regulates the allowable number of battery cycles per day. Specifically, it ensures that the cumulative LIB charging and discharging energy of all time steps (T_d) within each day (d) does not exceed the manufacturer's specified cycling limits (N_{Li}^{cyc}).

$$\sum_{t=(d-1)\cdot(T_d+1)}^{d\cdot T_d} \frac{\left[P_{Li}^{ch}(t)+P_{Li}^{dis}(t)\right]\cdot \Delta T}{2\times E_{Li}(t)} \leq N_{Li}^{cyc}, \quad \forall d \in \{1,2,\ldots,365\} \tag{14}$$

This constraint ensures that the battery operates within the recommended usage limits, thereby preventing any violation of its warranty conditions and preserving its useful life by minimising degradation.

Non-Negativity:

$$SoE_{Li}(t), SoE_{Li}(0), P_{Li}^{\mathrm{ch}}(t), P_{Li}^{\mathrm{dis}}(t), P_{ns}(t), P_{curt}(t), u_{Li}(t) \geq 0 \tag{15}$$

2.4.3. LIB degradation evaluation using BLAST (Model-Agnostic Framework)

The LIB degradation evaluation outlined in Eqs. (16)–(18) uses BLAST as the degradation engine. The proposed frameworks generate $SoC(t)$ and $T_{mp}(t)$, which are then used by the degradation engine to compute capacity loss.

$$L_{Li} = \frac{1-SoH_{Li}^{E}}{e_{loss}} \tag{16}$$

$$e_{loss,yr} = f_{BLAST}\left(SoC(t), T_{mp}(t)\right) \tag{17}$$

$$SoC(t) = \frac{SoE_{Li}(t)}{E_{Li}^{\max}} \tag{18}$$

Model-agnostic coupling: In this work, the battery operation optimisation produces the operating trajectories $SoC(t)$ and $T_{mp}(t)$. These trajectories are then passed to an external degradation function to compute capacity loss. Although BLAST is used in this paper, the proposed representative-day framework is not tied to BLAST. Any degradation model that estimates capacity fade from operating trajectories (e.g., SoC, temperature and time) can be used by replacing f_{BLAST} with another degradation function. For consistency and fair comparison, the same degradation model must be used for the Base Case and all benchmark methods.

Eq. (16) calculates the total battery life (L_{Li}) based on the allowable loss of battery capacity, defined as the difference between the initial capacity and the EoL threshold (SoH_{Li}^{E}), divided by the annual loss of energy capacity ($e_{loss,yr}$).

Eq. (17) defines $e_{loss,yr}$ over the simulation period, summing incremental degradation for each timestep. This incremental degradation is derived using the BLAST function, which accounts for SoC and battery temperature (T_{mp}) at each time step. The $SoC(t)$ used in BLAST is calculated using (18), representing $SoE_{Li}(t)$ divided by its E_{Li}^{max}.

2.4.4. Proposed frameworks algorithms

The Proposed-I framework (Algorithm 1) begins by initialising the simulation year (y), relative battery capacity ($RelCap_0$), and an empty dataset for storing cumulative results. The main iterative loop continues to run as long as the relative capacity of the LIB remains above the predefined EoL threshold SoH_{Li}^{E}. At the beginning of each simulated year, the available LIB capacity, $E_{Li}^{\max}(y)$, is updated according to the current relative capacity. Subsequently, an optimisation problem is individually solved for each representative day, using the predefined objective function ((1)) and operational constraints ((2)–(14)), with the battery size fixed at $E_{Li}^{\max}(y)$. After optimisation, daily results are replicated according to their weights (w_d) and assembled into an annual dataset, which is then merged into cumulative simulation data. Using this cumulative dataset, the BLAST-Lite model calculates annual battery degradation, $e_{\mathrm{loss,yr}}$, updating the relative capacity of the battery for the next year according to (19). The process repeats yearly, increasing the year counter, until the relative capacity falls below the EoL threshold, at which point the algorithm returns the annual degradation values and the total operational life (L_{Li}).

Algorithm 1: Proposed-I approach framework

```
Input:
    Representative day data set, {D_d}_{d=1}^n (P_load, P_solar, P_wind, T_mp),
    Weight of each representative day, w_d,
    Rated power, P_Li^max and energy, E_Li^max, capacity of the LIB
    End-of-life (EoL) threshold, SoH_Li^E (e.g. 0.8),
    Penalty factors, α_PF(t), β_PF(t), optimiser parameters, MIPgap,
    BLAST-Lite degradation model, f_BLAST(·)
Output:
    Yearly relative capacities, {RelCap_y}, and total LIB service life, L_Li
Set y ← 0, RelCap_0 ← 1.0, CumulativeResults ← ∅
while RelCap_y > SoH_Li^E do
    E_Li^max(y) ← E_Li^max × RelCap_y ; // available energy this year
    for d ← 1 to n do
        Solve daily MILP [objective (1), constraints (2)–(14)]
            with horizon T_d = 288,
            battery size fixed at E_Li^max(y)
            store DailyResults_d
    Assemble one-year data set
        𝒴_y ← [DailyResults_1^{×w_1}, …, DailyResults_n^{×w_n}]
    CumulativeResults ← CumulativeResults ∪ 𝒴_y
    Compute e_loss,yr from BLAST using (17) on CumulativeResults
    Update relative capacity using (19),

        RelCap_{y+1} = RelCap_y − e_loss,yr                    (19)
    y ← y + 1
return {e_loss,yr}, L_Li ← y;
```

The Proposed-II approach framework (Algorithm 2) provides a systematic method to evaluate battery degradation using a single-year simulation based on representative days. The algorithm begins by generating a time-dependent battery energy capacity profile ($E_{Li}(t)$) for the final year of operation, accounting for the battery's SoH decline from the start of the year to its end. For each representative day, the method extracts the load and renewable generation profiles and solves a daily MILP optimisation problem using the predefined objective ((1)) and constraints ((2)–(14)) considering the calculated energy capacity trajectory. The Optimisation results yield daily SoE profiles, which are replicated multiple times to expand each representative day into an extended dataset reflecting realistic operational conditions. The algorithm then calculates battery daily degradation by processing replicated SoC data and temperature profiles through the BLAST-Lite degradation model. The obtained daily degradation values are multiplied by their respective calendar-day weights to form cluster-weighted degradation measures. Aggregating these measures yields a representative annual degradation rate, from which the battery's remaining capacity at year-end is computed. Finally, this annual degradation rate is used to extrapolate and estimate the total battery service life. The algorithm concludes by returning comprehensive results, including the daily and annual degradation metrics, the terminal relative capacity, and the LIB's projected operational lifetime.

Algorithm 2: Proposed-II approach framework

Input:
Representative day data set, $\{\mathcal{D}_d\}_{d=1}^{n}$ (P_{load}, P_{solar}, P_{wind}, T_{mp}),
Weight of each representative day, w_d,
Rated power, $P_{Li}^{\max}$ and energy, $E_{Li}^{\max}$, capacity of the LIB
State-of-health at the start $\left(SoH_{Li}^{E-1}\right)$ and the end $\left(SoH_{Li}^{E}\right)$ of last operational year,
Penalty factors, $\alpha_{PF}(t)$, $\beta_{PF}(t)$, optimiser parameters, $MIPgap$
Replication factor, R (e.g. 100 days per representative day),
BLAST-Lite degradation model, $f_{\text{BLAST}}(\cdot)$
Output:
Daily degradation, $\{e_{\text{loss},d}\}$, cluster-weighted yearly degradation, $e_{\text{loss,yr}}^{\text{rep}}$,
Relative capacity at the end of single year, RelCap_E,
Compute time-varying capacity trajectory, $E_{Li}(t)$ for the last year via (11);
for $d \leftarrow 1$ **to** n **do**
  Extract $\mathcal{D}_d$ (288 timesteps);
  Solve daily MILP with objective (1) and constraints (2)–(14), using $E_{Li}(t)$;
  Store optimal profiles $\{SoE_{Li}(t)\}$;
  Replicate results R times $\rightarrow$ *ClusterData*$_d$;
for $d \leftarrow 1$ **to** n **do**
  Build SoC series $SoC(t) = SoE_{Li}(t)/E_{Li}^{\max}$ from *ClusterData*$_d$;
  Truncate temperature profile to length $T = 288R$;
  Run BLAST-Lite $f_{\text{BLAST}}\left(SoC(t), T_{mp}(t)\right)$;
  Compute daily degradation $e_{\text{loss},d}$ by (20);

$$e_{\text{loss},d} = \left(q_{T_d-1} - q_{T_d}\right) \tag{20}$$

  where q_{T_d} is the relative discharge capacity at the final timestep of representative day d;
  Compute cluster-weighted degradation $e_{\text{loss},d}^{cl} = w_d\, e_{\text{loss},d}$;
Aggregate yearly loss using (21);

$$e_{\text{loss,yr}}^{\text{rep}} = \sum_{d=1}^{n} w_d\, e_{\text{loss},d} \tag{21}$$

Update terminal relative capacity via (22);

$$\text{RelCap}_E = 1 - e_{\text{loss,yr}}^{\text{rep}} \tag{22}$$

Calculate life of LIB using extrapolation;

$$L_{Li} = \frac{1 - SoH_{Li}^{E}}{e_{\text{loss,yr}}^{\text{rep}}} \tag{23}$$

return $\{e_{\text{loss},d}\}$, $\{e_{\text{loss},d}^{cl}\}$, $e_{\text{loss,yr}}^{\text{rep}}$, RelCap_E, L_{Li};

2.5. Battery cost model

Estimation of battery degradation directly affects the EoL and dictates the battery lifetime, hence the replacement frequency throughout the duration of a project. Thus, a precise prediction of battery degradation is vital for accurately establishing the replacement cost schedule and the total NPC of the battery throughout the project's life. Our study includes a detailed cost analysis of battery replacement, operation, and salvage value, outlined in this section, as key components in evaluating economic viability. It includes the initial capital investment, ongoing operational and maintenance expenditures, replacement costs incurred throughout the battery's lifespan, and the salvage value at the project's conclusion.

The NPC of the LIB (C_N^{Li}) is expressed as:

$$C_N^{Li} = C_C^{Li} + C_O^{Li} + C_R^{Li} - C_S^{Li} \tag{24}$$

where C_C^{Li} is the total capital cost; C_O^{Li} is the total operational cost; C_R^{Li} is the total replacement cost and C_S^{Li} is the total salvage value. The total capital cost includes two main terms related to energy and power capacity, as follows [27]:

$$C_C^{Li} = C_e^{Li} + C_p^{Li} \tag{25}$$

$$C_e^{Li} = \sum_{e\in E} c_e^u \cdot E_{Li}^{max}; \forall k \in \{\texttt{ec}, \texttt{bs}, \texttt{si}, \texttt{ep}, \texttt{pd}\} \tag{26}$$

$$C_p^{Li} = \sum_{p\in P} c_p^u \cdot P_{Li}^{max}; \forall k \in \{\texttt{pc}, \texttt{cc}\} \tag{27}$$

where c_e^u unit cost associated with energy, including energy capacity (ec), balance of system (bs), system integration (si), engineering and procurement (ep), and project development (pd). Similarly, c_p^u is the unit cost associated with power, including the unit cost of the power conversion system (pc) and control and communication (cc). The operational cost of the LIB during the MG life can be expressed as:

$$C_O^{Li} = \sum_{y=1}^{L_{mg}} \frac{P_{Li}^{max} \cdot c_O^{u,Li}}{(1+i)^y} \tag{28}$$

where $c_O^{u,Li}$ is the unit O&M cost of the LIB.

The following Eqs. (29), (30), (31) present the replacement cost of the LIB (C_R^{Li}), which includes the replacement cost of power capacity (C_{Rp}^{Li}) and energy capacity (C_{Re}^{Li}).

$$C_R^{Li} = C_{Rp}^{Li} + C_{Re}^{Li} \tag{29}$$

$$C_{Rp}^{Li} = \sum_{y=1}^{\left\lfloor \frac{L_{mg}}{L_{\texttt{IN}}^{Li}} \right\rfloor} \frac{C_{\texttt{pc}}^{Li}}{(1+i)^{y\cdot L_{\texttt{IN}}^{Li}}} \tag{30}$$

$$C_{Re}^{Li} = \sum_{y=1}^{\left\lfloor \frac{L_{mg}}{L^{Li}} \right\rfloor} \frac{C_{\texttt{ec}}^{Li}}{(1+i)^{y\cdot L^{Li}}} \tag{31}$$

where L^{Li} is the life of the LIB batteries derived from the previous Section 2.4.3.

The salvage value models of LIB are not included in previous studies and are calculated as follows [27]:

$$C_S^{Li} = C_{Se}^{Li} + C_{Sp}^{Li} \tag{32}$$

$$C_{Se}^{Li} = C_e^{Li} \cdot \left(1 - \frac{2}{L^{Li}}\right)^{L_{us}^{Li}} \cdot d \tag{33}$$

$$C_{Sp}^{Li} = \left[C_{\texttt{pc}}^{Li} \cdot \left(1 - \frac{2}{L_{\texttt{IN}}^{Li}}\right)^{L_{us,\texttt{IN}}^{Li}} + C_{\texttt{bs}}^{Li} \cdot \left(1 - \frac{2}{L_{\texttt{bs}}^{Li}}\right)^{L_{mg}}\right] \cdot d \tag{34}$$

$$L_{us}^{Li} = L_{Li} - \left[\left(L_{Li} \times \left\lceil \frac{L_{mg}}{L_{Li}} \right\rceil\right) - L_{mg}\right] \tag{35}$$

$$L_{us,\texttt{IN}}^{Li} = L_{\texttt{IN}}^{Li} - \left[\left(L_{\texttt{IN}}^{Li} \times \left\lceil \frac{L_{mg}}{L_{\texttt{IN}}^{Li}} \right\rceil\right) - L_{mg}\right] \tag{36}$$

where C_{Se}^{Li} and C_{Sp}^{Li} are the salvage values of the energy and power capacities, respectively. L_{us}^{Li} is service life of the last battery pack; $L_{\texttt{IN}}^{Li}$ and $L_{us,\texttt{IN}}^{Li}$ are the battery inverter life and the service life of the last battery inverter, respectively. $L_{\texttt{bs}}^{Li}$ is the life of the balance of system components of the batteries.

2.6. Performance metrics

To systematically evaluate and compare the effectiveness of the conventional and proposed approaches, specific performance metrics are used. Primarily, the accuracy of battery degradation estimation is assessed, given its significant influence on battery replacement timelines within the project lifespan, directly impacting the NPC. Accurate degradation estimation ensures reliable scheduling of battery replacements. The accuracy of all approaches is validated by comparing them with the detailed Base Case scenario, ensuring robustness and reliability of the proposed methodologies. The discrepancy in accuracy for each method compared to that year's baseline net profile is computed as:

$$A_{Dif} = \frac{AD_{approach} - AD_{base}}{AD_{base}} \times 100\% \tag{37}$$

where A_{Dif} is the percentage accuracy difference, $AD_{approach}$ is the annual degradation of the approach and AD_{base} is the annual degradation of the base case. In addition, the computational efficiency of solving battery degradation estimation, measured as the computational time in minutes, is considered to determine the practical feasibility of each approach.

3. Simulation case study of a renewable microgrid

3.1. Simulation setup and implementation details: A case study in a renewable microgrid

The case study considered in this work involves a fully-renewable industrial microgrid system equipped with a solar PV system, a wind turbine, and LIB storage. This microgrid is designed to meet the energy requirements of an industrial load whose demand profile has been collected at 5-minute intervals over one year (105,120 timesteps). The battery operation framework, which is an optimisation problem, has been formulated in Section 2.4. Normalised solar and wind power generation profiles were derived based on site-specific solar irradiance and wind speed data, respectively; details can be found in the supplementary file [28]. Considering the type of industrial load, demand is not expected to change significantly from year to year; therefore, a year of load data captures all relevant variations. Although the input data for operational load demand covers only one year, the simulation framework allows repeated cycling of this yearly profile to simulate microgrid operation continuously until the battery reaches its EoL threshold.

To reduce the computational time in simulating battery operation, the full year's dataset can be effectively condensed into a smaller, representative subset of days. As discussed in Section 2.1, among several available clustering techniques, K-means is widely recognised for its reliability and computational efficiency and was therefore chosen for this study. K-means method is used to group the complete annual dataset (365 days) into a manageable set of five representative days, significantly streamlining the simulation process. Each of these representative days is assigned a weight that corresponds to the frequency of occurrence of similar load and generation patterns throughout the year. Note that the proposed frameworks are not limited to K-means or to five representative days, as any other clustering method could be employed. A sensitivity analysis on the number of representative days was also performed to evaluate the trade-off between computational burden and accuracy. The exact number of representative days can be adjusted based on specific accuracy requirements and the case study.

3.2. Input parameters

The results presented in this section are derived from a case study based on an existing underground mine in Australia. The case study assumes a LIB with a maximum power output ($P_{Li}^{\max}$) of 12 MW and an energy capacity ($E_{Li}^{\max}$) of 6 MWh. Both solar PV ($P_{solar}^{\max}$) and wind turbine systems ($P_{wind}^{\max}$) have been assigned an installed capacity of 12 MW each, and their respective real-time contributions are derived from site-specific irradiance and wind speed measurements from 2014 to 2023. A uniform cell temperature of 25 °C was imposed at every time step in all simulated years, thus eliminating interannual thermal variability and allowing an equitable comparison of degradation outcomes. This is justified as battery thermal management systems maintain the cell temperature within a tight range. To account for the importance of minimising power not served (P_{ns}) and renewable curtailment (P_{curt}), two penalty factors, i.e., α_{PF} and β_{PF}, are used. The former (β_{PF}) starts at a high value (1000) and gradually decreases to a lower value (1), emphasising the need to reduce the power not served more aggressively in earlier timesteps; the latter (α_{PF}) decreases from 1 to 0.001, penalising renewable power curtailment less over time. Accordingly, the LIB C-rate is fixed at 2C in this study, which aligns with values commonly reported in the literature [29–31], indicating that it can charge or discharge twice its energy capacity per hour; notably, higher C-rates have also been demonstrated in practice, such as the Dalrymple Battery Energy Storage System in South Australia, where operation up to 3.75C has been reported [32]. Further, 95% inverter efficiency (η_{Inv}) [33] and 95% LIB charging (η_{Li}^{ch}) and discharging (η_{Li}^{dis}) efficiency [34] are assumed. The minimum and maximum SoE ratios ($\alpha_{SoE}^{\min}$ and $\alpha_{SoE}^{\max}$) are 0.1 and 0.9, respectively, to ensure that the battery never operates in regions of excessively deep discharge or overcharge. Considering a nominal SoH of 0.82 (SoH_{Li}^{E-1}) at the beginning of the last year of operation and an EoL threshold of 0.80 (SoH_{Li}^{E}), the system allows only one complete cycle per day (N_{Li}^{cyc}=1) to limit the accelerated degradation [35,36]. These parameters were chosen to reflect realistic battery operation constraints for an underground mine in Australia. Detailed cost assumptions and economic parameters are discussed further in the supplementary document [28].

3.3. Simulation tools and environment

All simulation and optimisation tasks within this study have been implemented using Python. Specifically, battery degradation analyses use BLAST, a Python-based package developed by NREL [37], to accurately quantify battery ageing considering cyclic and calendar degradation phenomena. To solve the battery operation optimisation problems formulated in this research, the Gurobi solver [38] (Academic Licence) is employed. All simulations were executed locally on a 64-bit Windows laptop equipped with an 11th-Generation Intel® Core™ i7-1165G7 processor (base 2.80 GHz) and 16 GB of RAM.

4. Results and discussions

4.1. Battery operation and degradation analysis using different approaches

The results of the proposed and conventional approaches, along with the base case framework, have been derived using the annual profile of renewable generation for 2014–2023 (10 years). The Base Case, Conventional I, and Proposed I approaches are designed to comprehensively analyse LIB degradation patterns throughout the battery's operational life until it reaches its defined EoL threshold. In contrast, single-year operational scenarios, including Conventional-II, III, IV, and Proposed-II, estimate battery lifetime by extrapolating from annual degradation results obtained through simulations over one year. Specifically, the Proposed-II approach additionally provides details on the selected representative days, including their individual cluster degradation. For those approaches which involve representative days, five representative days have been chosen to characterise the whole year. Furthermore, computational times have been recorded for all approaches, allowing for a comparative assessment of their computational efficiency.

Fig. 4 compares the seven degradation estimation approaches across ten renewable annual profiles (2014 to 2023). In all years of the annual

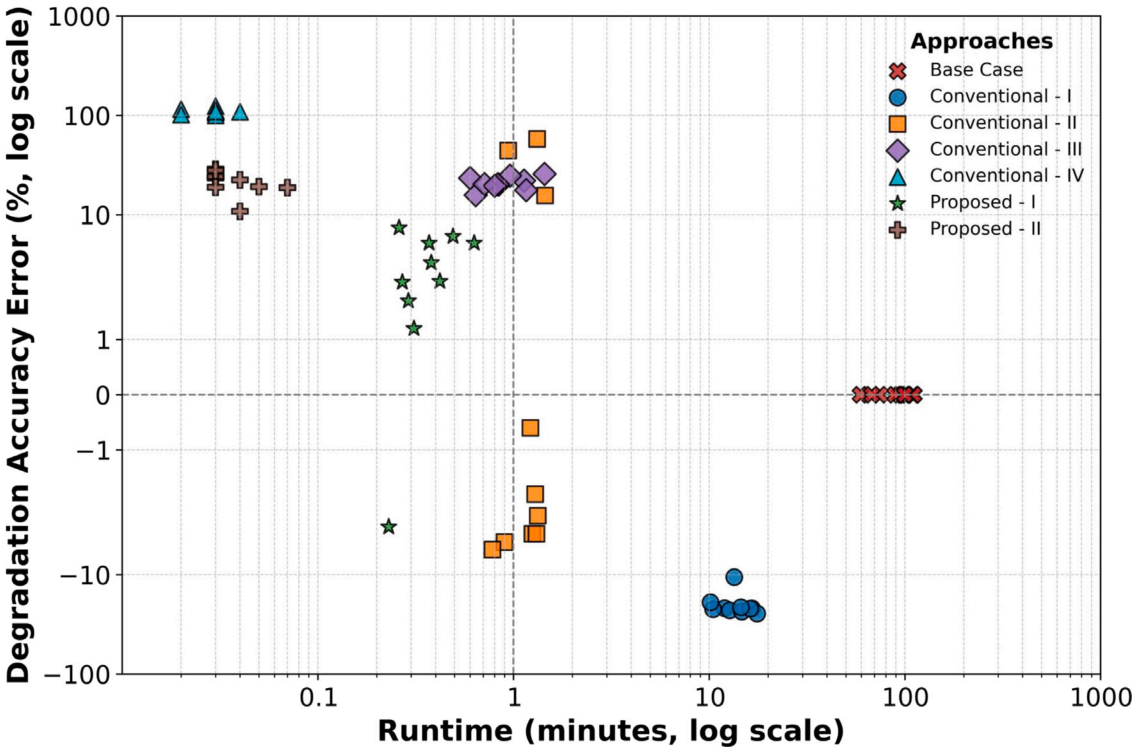


Fig. 4. Comparison of LIB degradation estimation accuracy and runtime across different approaches in ten years of simulation.

renewable generation profile, the Base Case scenario involves detailed day-by-day battery simulations until the EoL threshold, which accurately captures average annual battery degradation but requires considerable computational effort, approximately between 59 and 111 min for a single run. None of the conventional approaches reproduces the “ground truth” in the Base Case within an acceptable error. Conventional-II delivers the smallest deviations (≈ 2%–9% below Base Case in most years, with an average error of ≈ 14%) but requires 1-2 min per run, three times slower than the cluster-based methods, which could be significant in the planning optimisation problem. At the opposite extreme, Conventional-IV finishes in only ≈ 0.03 min; however, its degradation error increases significantly to 100%–124%, which is clearly unacceptable.

The two proposed schemes sit between these extremes. Proposed-I limits the error band to 1%–8% (with an average error of 3.76%) while keeping the runtime between 0.2–0.7 min, and the ultrafast approach Proposed-II offers the absolute shortest runtimes (0.02–0.04 min) with most annual renewable generation profiles having moderate, year-to-year consistent error levels (with an average error of 21.84%). Although all methods incur inaccuracies, the proposed approaches offer the most stable balance between computational efficiency and degradation fidelity, making them practical surrogates when the exact Base Case simulation is computationally prohibitive. To further analyse and clarify the outcomes presented in Fig. 4, Table A.1 in the appendix provides a comprehensive comparison across all the annual profiles of renewable generation analysed (2014–2023), highlighting the LIB life estimation of each approach relative to their respective base cases, as well as their average annual battery degradation.

4.2. Detailed discussions of the proposed approaches

Among the ten annual renewable-generation profiles evaluated (Appendix Table A.1), 2016 and 2021 years have been discussed as example cases for Proposed-I to demonstrate how the results should be read and compared against the corresponding Base Case. For the 2016 year, the Proposed-I approach completed the multiyear degradation assessment in merely 0.29 min, approximately four times faster than the Conventional-II approach, which required 1.25 min. Despite this significant reduction in runtime, Proposed-I maintained an accuracy within | 1.7% | relative to the Base Case. Similarly, in 2021, the Proposed-I method completed the assessment in just 0.31 min, approximately four times faster than the Conventional-II’s 1.22 min, while preserving accuracy within | 1.2% | compared to the 2021 Base Case. These results underscore the efficacy and accuracy of the Proposed-I approach in conducting rapid and reliable multiyear degradation assessments.

For completeness, Proposed-II is also elaborated using two example years (2020 and 2022), while the full ten-year results are provided in Table A.1. The Proposed-II approach significantly reduces computational runtime, achieving near-instantaneous results. For they 2020 year, Proposed-II completes the multiyear degradation simulation in merely 0.04 min, accepting a slightly larger deviation of | 10.8% | from the corresponding Base Case degradation value. Similarly, for 2022, which emerges as the second-best performing year for this approach, Proposed-II finalises the simulation in just 0.07 min, resulting in an error margin of | 18.7% |. Although these errors appear relatively high, they represent the lowest achievable error than the existing ultrafast method Conventional-IV (average error more than 100%), highlighting their advantage in terms of computational efficiency compared to other available approaches. Together, these results show that the proposed approaches achieve the best overall compromise.

To provide a deeper understanding of the detailed degradation patterns for one of the optimal scenarios, a graphical analysis for 2021 is presented in Fig. 5. This figure comprehensively depicts the battery’s relative capacity degradation trajectory throughout its operational life until reaching the EoL threshold. In this figure, the Base Case, Proposed–I, and Conventional–I approaches clearly illustrate the LIB degradation trajectories throughout the operational lifespan up to the EoL threshold, represented by the solid lines. Conversely, the Conventional–II, III, IV, and Proposed–II approaches depict degradation trajectories based on only a single year of battery operation (solid lines), and then extrapolated these results to estimate the battery EoL

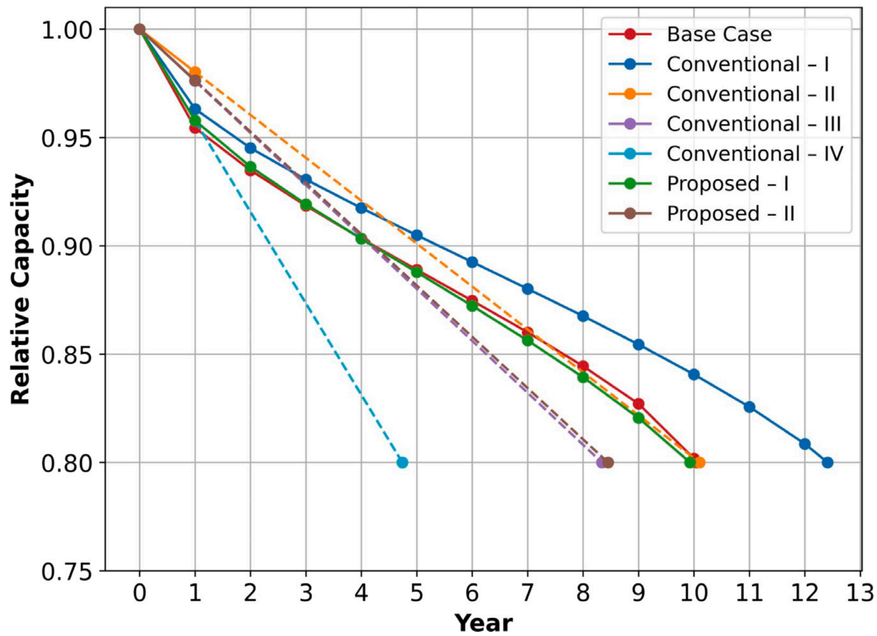


Fig. 5. Comprehensive LIB degradation using different approaches in year 2021.

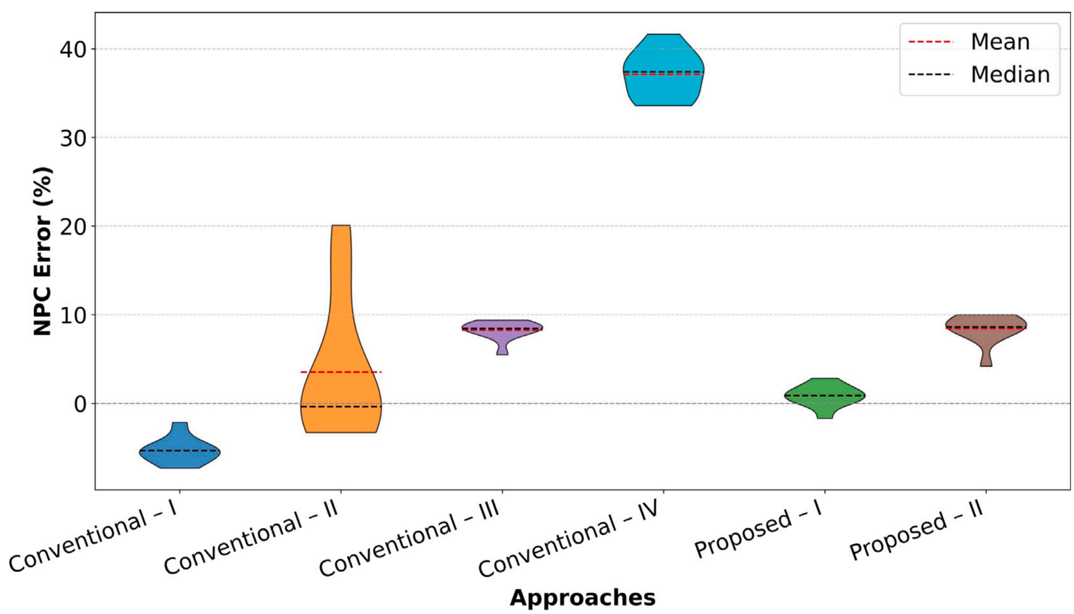


Fig. 6. NPC error distribution by each approach compared to the Base Case.

using dashed lines. In summary, Proposed-I and Conventional-II most accurately replicate the actual trajectory for that year where Proposed-I is much consistent and faster.

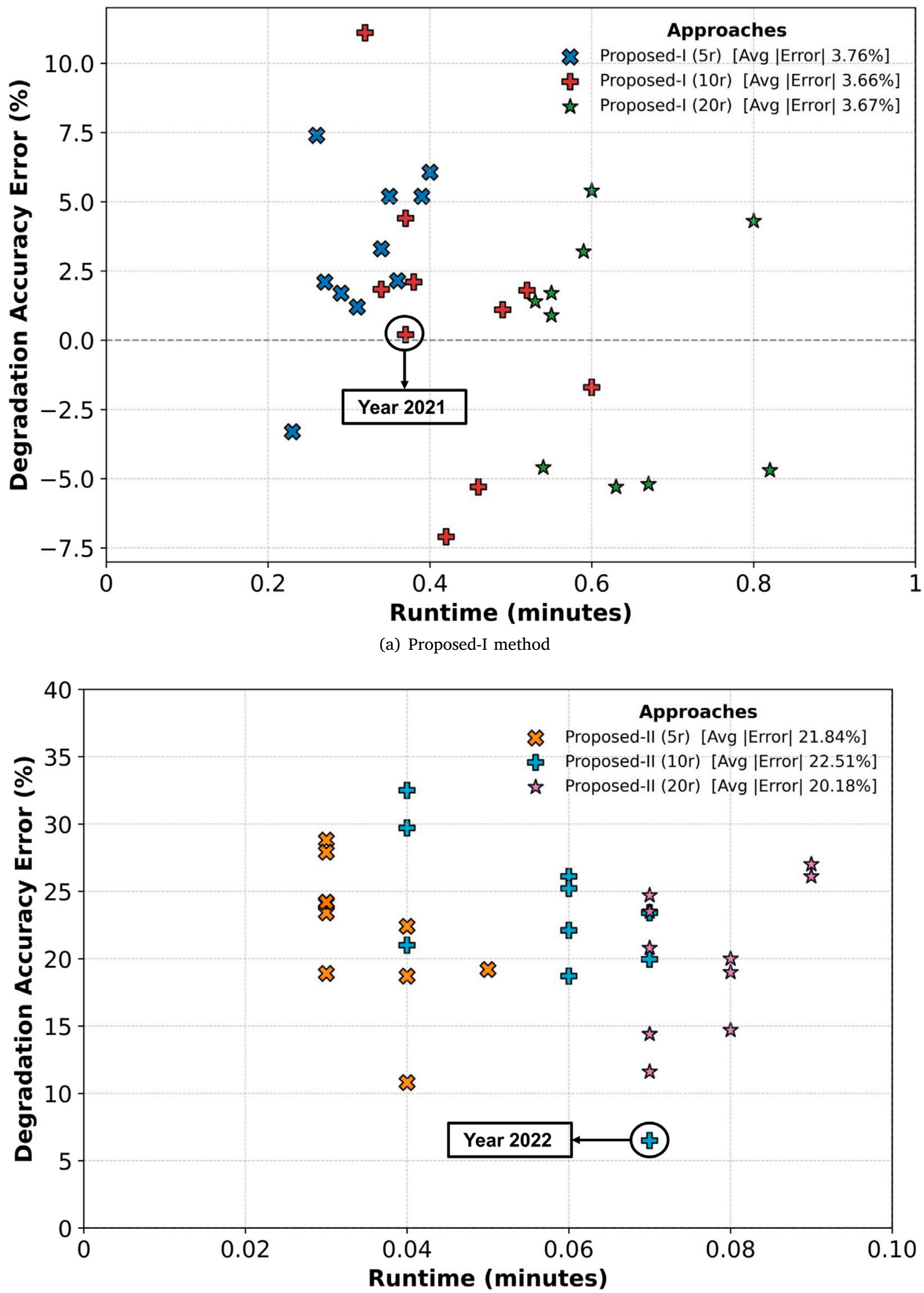
4.3. Cost analysis

The cost models have been applied to all years (2014–2023) to determine the LIB NPC over a 30-year project lifetime. Battery degradation significantly impacts the number of replacements required within the project's duration, as even minor differences in annual degradation rates can lead to notable variations in battery replacement schedules, associated replacement costs, and salvage values at the project's conclusion. In contrast, other cost components, such as initial capital investment and operational expenditures, remain consistent across all evaluated approaches. Table A.2 summarises the NPC calculated for each approach using ten years of data, and the NPC error with respect to the Base Case for each year (illustrated in Fig. 6). The results clearly show that the Proposed–I approach achieves the lowest NPC error in all evaluated years, ranging from only 0.25% to 2.84%, with both its mean and median errors remaining below 1%. This consistently high level of accuracy underscores the practical viability of the Proposed–I approach in cost-sensitive planning studies.

4.4. Sensitivity analyses

4.4.1. Number of representative days

A sensitivity analysis was performed on Proposed–I and Proposed-II methods using a decade of data to assess how different numbers of representative days affect battery degradation accuracy and computational efficiency. The original Proposed–I and Proposed-II approaches employ



(a) Proposed-I method

(b) Proposed-II method

Fig. 7. Sensitivity analyses of LIB degradation estimation accuracy and runtime for ten years with increased number of representative days for the Proposed-I and Proposed-II approaches.

five representative days to approximate the entire year's operational profile. To examine the potential improvement or deterioration in degradation prediction accuracy relative to the base case scenario, two additional variations were implemented using 10 representative days, namely Proposed–I(10r) and Proposed-II(10r), and 20 representative days, namely Proposed-I(20r) and Proposed-II(20r).

The simulation results are shown in Fig. 7. Although increasing the number of representative days could potentially improve the accuracy of the degradation estimates, improvements in accuracy are not guaranteed. This is because representative days are obtained using K-means centroids, which minimise the average within-cluster squared error and therefore represent the most frequent (typical) operating

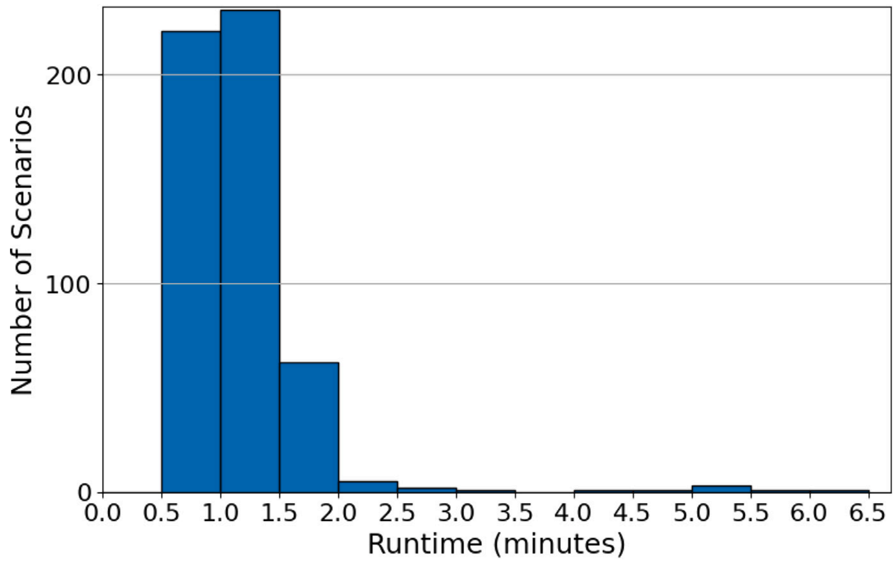


Fig. 8. Effect of LIB configurations (500 scenarios) impact on Proposed - I approach's simulation runtime, with power capacity range from 4MW to 12MW and C-rate range from 0.1 to 2.

patterns most accurately. As a result, increasing the number of representative days from 5 to 10 or 20 may mainly subdivide the dense region of common days, while rare but degradation-dominant extreme days (e.g., renewable drought or very high net-demand days) can still be underrepresented or smoothed by centroid averaging. Since battery degradation responds non-linearly to these stress events, improved profile matching does not necessarily translate into improved degradation accuracy. Similar findings have been reported in time-series aggregation studies: centroid-based typical periods can bias key planning outcomes by smoothing extremes, and increasing the number of typical periods does not necessarily improve accuracy for nonlinear metrics unless extreme periods are explicitly represented [39,40]. Additionally, incorporating additional representative days moderately increased computational runtime. Figs. 7(a) and 7(b) illustrate that the mean absolute error remained largely unchanged. The detailed results are summarised in Tables A.3 and A.4 of Appendix.

The Base Case approach, while precise, becomes impractical for battery sizing and long-term planning studies due to the potentially long runtimes for a single sizing study. In contrast, the proposed-I approach significantly streamlines computational efficiency by using a handful of representative days to approximate annual operations. Among the sensitivity analyses of the Proposed-I approach in Fig. 7(a), Proposed-I(10r) achieves the lowest average error of the degradation estimate of 3.66%, and among all years. 2021 stood out to be the best, which estimates the error within only 0.2% of the Base Case while substantially reducing the runtime to only 0.37 min, representing a nearly two hundredfold improvement compared to the Base Case.

Moreover, the Proposed–II approach's sensitivity analyses deliver a dramatic improvement in computational efficiency, completing the simulation in 0.02 to 0.09 min, a thousandfold enhancement over the Base Case approach. Among the sensitivity analyses of Proposed-II approach in Fig. 7(b), Proposed-II(10r) for the year 2022 achieves the lowest error of the degradation estimate of 6.5%, while reducing the runtime to only 0.07 min. These substantial improvements in computational speed and reliability make the ultrafast approach particularly attractive for industry practitioners who require rapid battery planning analyses, trading a modest increase in degradation estimation error for significantly shorter simulation times compared to the Proposed–I approach.

4.4.2. Runtime analysis of different LIB configurations

In practice, many battery developers and consultants adopt an early-stage scenario-based evaluation — rather than embedding the sizing decision directly inside an optimisation routine — to screen a broad set of capacity–C-rate pairs and shortlist promising designs [28]. In this context, the solver must be run repeatedly, and its runtime is sensitive to the selected battery size: larger energy capacities and their corresponding power ratings can either increase — or occasionally decrease — the combined optimisation and degradation loops. To quantify this effect, the following subsection measures how computation time scales when LIB capacity and C-rate are varied across the scenario space, thereby illustrating the practical runtime implications of using our accelerated degradation frameworks within an NPC-oriented planning study. Ideally, the runtime variations across different capacities and C-rates should remain minimal, preferably the majority of the simulations should take about 1 min, to maintain the efficiency of extensive planning and techno-economic analyses. To verify this aspect, multiple LIB sizes and C-rates have been tested within the Proposed-I(10r) operational simulation framework. Between Proposed-I and Proposed-II, Proposed-I(10r) has been chosen because it has a longer simulation time than Proposed-II.

The results of the sensitivity analysis are presented in Fig. 8, which illustrates runtime variations across a set of 500 scenarios of Proposed-I. These scenarios cover LIB power capacities ranging from 4 MW to 12 MW and C-rates varying between 0.1 and 2. Most of these simulations consistently achieve a computational runtime of less than one minute, highlighting the robustness and computational efficiency of the Proposed–I(10r) framework in a wide variety of battery configurations and operational scenarios.

4.5. Summary

The results presented in this section demonstrate that the Proposed–I approach offers a robust balance between computational efficiency and accuracy in battery degradation and economic evaluations. Sensitivity analyses confirm that increasing the number of representative days further enhances accuracy, achieving near-Base Case precision at a substantially reduced computational runtime. Additionally, runtime analyses across diverse LIB configurations show consistent efficiency,

Table A.1
LIB degradation summary across multiple annual renewable generation profiles.

Year	Approach	Annual degradation	Runtime	EoL (SoH=0.80)	Difference (+/- than Base)
2014	Base Case	1.878%	109.62 min	10.65 years	N/A
	Conventional – I	1.469%	16.55 min	13.61 years	(-) 21.8%
	Conventional – II	1.830%	1.33 min	10.92 years	(-) 2.56%
	Conventional – III	2.293%	1.13 min	8.72 years	(+) 22.1%
	Conventional – IV	3.884%	0.03 min	5.15 years	(+) 106.8%
	Proposed – I	1.992%	0.49 min	10.04 years	(+) 6.07%
	Proposed – II	2.419%	0.03 min	8.26 years	(+) 28.8%
2015	Base Case	1.890%	98.75 min	10.59 years	N/A
	Conventional – I	1.426%	17.59 min	14.017 years	(-) 24.6%
	Conventional – II	2.185%	1.45 min	9.153 years	(+) 15.6%
	Conventional – III	2.378%	1.44 min	8.41 years	(+) 25.8%
	Conventional – IV	3.949%	0.03 min	5.064 years	(+) 108.9%
	Proposed – I	1.989%	0.63 min	10.054 years	(+) 5.2%
	Proposed – II	2.313%	0.04 min	8.65 years	(+) 22.4%
2016	Base Case	1.962%	94.24 min	10.19 years	N/A
	Conventional – I	1.536%	11.97 min	13.02 years	(-) 21.7%
	Conventional – II	1.886%	1.25 min	10.60 years	(-) 3.9%
	Conventional – III	2.358%	0.84 min	8.48 years	(+) 20.2%
	Conventional – IV	3.934%	0.03 min	5.08 years	(+) 100.5%
	Proposed – I	1.996%	0.29 min	10.02 years	(+) 1.7%
	Proposed – II	2.434%	0.03 min	8.21 years	(+) 24.1%
2017	Base Case	1.905%	111.03 min	10.50 years	N/A
	Conventional – I	1.471%	12.73 min	13.59 years	(-) 22.8%
	Conventional – II	3.009%	1.32 min	6.64 years	(+) 58.0%
	Conventional – III	2.375%	0.96 min	8.42 years	(+) 24.7%
	Conventional – IV	4.105%	0.02 min	4.87 years	(+) 115.5%
	Proposed – I	2.004%	0.37 min	9.98 years	(+) 5.2%
	Proposed – II	2.350%	0.03 min	8.51 years	(+) 23.4%
2018	Base Case	1.968%	74.40 min	10.16 years	N/A
	Conventional – I	1.530%	10.48 min	13.07 years	(-) 22.3%
	Conventional – II	1.876%	0.90 min	10.66 years	(-) 4.7%
	Conventional – III	2.367%	0.83 min	8.45 years	(+) 20.3%
	Conventional – IV	4.134%	0.03 min	4.83 years	(+) 110.1%
	Proposed – I	2.032%	0.38 min	9.84 years	(+) 3.3%
	Proposed – II	2.445%	0.03 min	8.18 years	(+) 24.2%
2019	Base Case	1.904%	88.77 min	10.51 years	N/A
	Conventional – I	1.702%	13.43 min	11.75 years	(-) 10.6%
	Conventional – II	2.743%	0.94 min	7.29 years	(+) 44.1%
	Conventional – III	2.350%	0.60 min	8.51 years	(+) 23.4%
	Conventional – IV	4.256%	0.03 min	4.69 years	(+) 123.5%
	Proposed – I	2.045%	0.26 min	9.78 years	(+) 7.4%
	Proposed – II	2.435%	0.03 min	8.21 years	(+) 27.9%
2020	Base Case	2.094%	59.31 min	9.57 years	N/A
	Conventional – I	1.603%	14.66 min	12.47 years	(-) 23.5%
	Conventional – II	1.977%	0.78 min	10.11 years	(-) 5.6%
	Conventional – III	2.424%	0.64 min	8.25 years	(+) 15.8%
	Conventional – IV	4.226%	0.02 min	4.73 years	(+) 101.9%
	Proposed – I	2.024%	0.23 min	9.88 years	(-) 3.3%
	Proposed – II	2.320%	0.04 min	8.62 years	(+) 10.8%
2021	Base Case	1.990%	67.69 min	10.05 years	N/A
	Conventional – I	1.611%	10.16 min	12.41 years	(-) 19.0%
	Conventional – II	1.979%	1.22 min	10.10 years	(-) 0.6%
	Conventional – III	2.396%	0.71 min	8.34 years	(+) 20.4%
	Conventional – IV	4.219%	0.03 min	4.74 years	(+) 112.1%
	Proposed – I	2.014%	0.31 min	9.93 years	(+) 1.2%
	Proposed – II	2.367%	0.03 min	8.45 years	(+) 18.9%
2022	Base Case	1.957%	100.92 min	10.22 years	N/A
	Conventional – I	1.529%	16.22 min	13.07 years	(-) 21.9%
	Conventional – II	1.881%	1.31 min	10.63 years	(-) 3.9%
	Conventional – III	2.303%	1.16 min	8.68 years	(+) 17.7%
	Conventional – IV	4.098%	0.03 min	4.88 years	(+) 109.4%
	Proposed – I	1.999%	0.42 min	10.001 years	(+) 2.15%
	Proposed – II	2.322%	0.07 min	8.61 years	(+) 18.7%
2023	Base Case	1.957%	99.81 min	10.22 years	N/A
	Conventional – I	1.540%	14.51 min	12.98 years	(-) 21.3%
	Conventional – II	1.922%	1.29 min	10.40 years	(-) 1.8%
	Conventional – III	2.341%	0.80 min	8.54 years	(+) 19.6%
	Conventional – IV	4.078%	0.04 min	4.90 years	(+) 108.4%
	Proposed – I	1.998%	0.27 min	10.01 years	(+) 2.1%
	Proposed – II	2.333%	0.05 min	8.57 years	(+) 19.2%

Table A.2
LIB net present cost (NPC) summary across multiple annual renewable-generation profiles.

Year	Approach	EoL (SoH = 0.80)	NPC (30 years project)	Difference (+/− than Base)
2014	Base Case	10.65 years	11.831 M$	N/A
	Conventional – I	13.61 years	11.133 M$	(-) 5.9%
	Conventional – II	10.92 years	11.778 M$	(-) 0.45%
	Conventional – III	8.72 years	12.810 M$	(+) 8.3%
	Conventional – IV	5.15 years	15.868 M$	(+) 34.1%
	Proposed – I	10.04 years	11.943 M$	(+) 0.95%
	Proposed – II	8.26 years	13.015 M$	(+) 10.0%
2015	Base Case	10.59 years	11.843 M$	N/A
	Conventional – I	14.017 years	11.016 M$	(-) 6.99%
	Conventional – II	9.153 years	12.585 M$	(+) 6.3%
	Conventional – III	8.41 years	12.951 M$	(+) 9.4%
	Conventional – IV	5.064 years	15.938 M$	(+) 34.6%
	Proposed – I	10.054 years	11.941 M$	(+) 0.83%
	Proposed – II	8.65 years	12.843 M$	(+) 8.44%
2016	Base Case	10.19 years	11.917 M$	N/A
	Conventional – I	13.02 years	11.294 M$	(-) 5.23%
	Conventional – II	10.60 years	11.841 M$	(-) 0.64%
	Conventional – III	8.48 years	12.921 M$	(+) 8.4%
	Conventional – IV	5.08 years	15.925 M$	(+) 33.6%
	Proposed – I	10.02 years	11.947 M$	(+) 0.25%
	Proposed – II	8.21 years	13.035 M$	(+) 9.4%
2017	Base Case	10.50 years	11.860 M$	N/A
	Conventional – I	13.59 years	11.139 M$	(-) 6.1%
	Conventional – II	6.64 years	14.243 M$	(+) 20.1%
	Conventional – III	8.42 years	12.947 M$	(+) 9.2%
	Conventional – IV	4.87 years	16.419 M$	(+) 38.5%
	Proposed – I	9.98 years	12.053 M$	(+) 1.6%
	Proposed – II	8.51 years	12.907 M$	(+) 8.8%
2018	Base Case	10.16 years	11.922 M$	N/A
	Conventional – I	13.07 years	11.281 M$	(-) 5.38%
	Conventional – II	10.66 years	11.829 M$	(-) 0.78%
	Conventional – III	8.45 years	12.934 M$	(+) 8.5%
	Conventional – IV	4.83 years	16.515 M$	(+) 38.5%
	Proposed – I	9.84 years	12.154 M$	(+) 1.95%
	Proposed – II	8.18 years	13.047 M$	(+) 9.4%
2019	Base Case	10.51 years	11.858 M$	N/A
	Conventional – I	11.75 years	11.603 M$	(-) 2.15%
	Conventional – II	7.29 years	13.636 M$	(+) 15.0%
	Conventional – III	8.51 years	12.907 M$	(+) 8.85%
	Conventional – IV	4.69 years	16.800 M$	(+) 41.66%
	Proposed – I	9.78 years	12.195 M$	(+) 2.84%
	Proposed – II	8.21 years	13.035 M$	(+) 9.93%
2020	Base Case	9.57 years	12.335 M$	N/A
	Conventional – I	12.47 years	11.434 M$	(-) 7.3%
	Conventional – II	10.11 years	11.931 M$	(-) 3.3%
	Conventional – III	8.25 years	13.019 M$	(+) 5.5%
	Conventional – IV	4.73 years	16.726 M$	(+) 35.6%
	Proposed – I	9.88 years	12.125 M$	(-) 1.7%
	Proposed – II	8.62 years	12.857 M$	(+) 4.2%
2021	Base Case	10.05 years	11.941 M$	N/A
	Conventional – I	12.41 years	11.449 M$	(-) 4.12%
	Conventional – II	10.10 years	11.933 M$	(-) 0.07%
	Conventional – III	8.34 years	12.981 M$	(+) 8.70%
	Conventional – IV	4.74 years	16.706 M$	(+) 39.9%
	Proposed – I	9.93 years	12.089 M$	(+) 1.24%
	Proposed – II	8.45 years	12.934 M$	(+) 8.32%
2022	Base Case	10.22 years	11.911 M$	N/A
	Conventional – I	13.07 years	11.281 M$	(-) 5.3%
	Conventional – II	10.63 years	11.835 M$	(-) 0.6%
	Conventional – III	8.68 years	12.829 M$	(+) 7.7%
	Conventional – IV	4.88 years	16.394 M$	(+) 37.6%
	Proposed – I	10.001 years	11.950 M$	(+) 0.3%
	Proposed – II	8.61 years	12.862 M$	(+) 8.0%
2023	Base Case	10.22 years	11.911 M$	N/A
	Conventional – I	12.98 years	11.305 M$	(-) 5.09%
	Conventional – II	10.40 years	11.878 M$	(-) 0.28%
	Conventional – III	8.54 years	12.894 M$	(+) 8.25%
	Conventional – IV	4.90 years	16.342 M$	(+) 37.2%
	Proposed – I	10.01 years	11.949 M$	(+) 0.32%
	Proposed – II	8.57 years	12.880 M$	(+) 8.14%

Table A.3
Sensitivity analysis of Proposed-I approaches with increased numbers of representative days.

Year	Approach	Annual degradation	Runtime	EoL (SoH=0.80)	Difference (+/– than Base)
2014	Base Case	1.878%	109.62 min	10.65 years	N/A
	Proposed – I (5r)	1.992%	0.40 min	10.04 years	(+) 6.07%
	Proposed – I (10r)	1.899%	0.49 min	10.53 years	(+) 1.1%
	Proposed – I (20r)	1.980%	0.60 min	10.10 years	(+) 5.4%
2015	Base Case	1.890%	98.75 min	10.59 years	N/A
	Proposed – I (5r)	1.989%	0.39 min	10.054 years	(+) 5.2%
	Proposed – I (10r)	1.858%	0.60 min	10.76 years	(–) 1.7%
	Proposed – I (20r)	1.790%	0.63 min	11.17 years	(–) 5.3%
2016	Base Case	1.962%	94.24 min	10.19 years	N/A
	Proposed – I (5r)	1.996%	0.29 min	10.02 years	(+) 1.7%
	Proposed – I (10r)	1.998%	0.34 min	10.01 years	(+) 1.84%
	Proposed – I (20r)	1.990%	0.53 min	10.05 years	(+) 1.4%
2017	Base Case	1.905%	111.03 min	10.50 years	N/A
	Proposed – I (5r)	2.004%	0.35 min	9.98 years	(+) 5.2%
	Proposed – I (10r)	1.988%	0.37 min	10.06 years	(+) 4.4%
	Proposed – I (20r)	1.966%	0.59 min	10.17 years	(+) 3.2%
2018	Base Case	1.968%	74.40 min	10.16 years	N/A
	Proposed – I (5r)	2.032%	0.34 min	9.84 years	(+) 3.3%
	Proposed – I (10r)	2.010%	0.38 min	9.95 years	(+) 2.1%
	Proposed – I (20r)	1.986%	0.55 min	10.07 years	(+) 0.90%
2019	Base Case	1.904%	88.77 min	10.51 years	N/A
	Proposed – I (5r)	2.045%	0.26 min	9.78 years	(+) 7.4%
	Proposed – I (10r)	2.116%	0.32 min	9.45 years	(+) 11.1%
	Proposed – I (20r)	1.986%	0.80 min	10.07 years	(+) 4.3%
2020	Base Case	2.094%	59.31 min	9.57 years	N/A
	Proposed – I (5r)	2.024%	0.23 min	9.88 years	(–) 3.3%
	Proposed – I (10r)	1.982%	0.46 min	10.09 years	(–) 5.3%
	Proposed – I (20r)	1.996%	0.82 min	10.02 years	(–) 4.7%
2021	Base Case	1.990%	67.69 min	10.05 years	N/A
	Proposed – I (5r)	2.014%	0.31 min	9.93 years	(+) 1.2%
	Proposed – I (10r)	1.994%	0.37 min	10.03 years	(+) 0.20%
	Proposed – I (20r)	1.899%	0.54 min	10.53 years	(–) 4.6%
2022	Base Case	1.957%	100.92 min	10.22 years	N/A
	Proposed – I (5r)	1.999%	0.36 min	10.001 years	(+) 2.15%
	Proposed – I (10r)	1.819%	0.42 min	10.99 years	(–) 7.1%
	Proposed – I (20r)	1.855%	0.67 min	10.78 years	(–) 5.2%
2023	Base Case	1.957%	99.81 min	10.22 years	N/A
	Proposed – I (5r)	1.998%	0.27 min	10.01 years	(+) 2.1%
	Proposed – I (10r)	1.992%	0.52 min	10.04 years	(+) 1.8%
	Proposed – I (20r)	1.990%	0.55 min	10.05 years	(+) 1.7%

enabling practical application for extensive techno-economic evaluations. Thus, the Proposed–I framework emerges as a practical choice for practical LIB sizing, planning, and techno-economic analysis.

5. Conclusions

This study developed two novel frameworks based on the representative-day concept that estimate long-term LIB degradation with manageable computational requirements needed for sizing and planning studies. Each framework was rigorously benchmarked against all established approaches along with a detailed day-to-day "ground-truth" simulation, demonstrating that the proposed methods deliver the best balance between computational efficiency and accuracy for practical battery planning applications. In particular, the first proposed framework (Proposed-I approach) accurately captures long-term LIB degradation until the EoL through strategic day replication and annual capacity updates, leading to small deviations in EoL predictions and cost assessments compared to more exhaustive day-to-day simulations.

In addition to showing how our two representative-day frameworks can closely replicate detailed day-to-day degradation simulations at a fraction of the computational time, we performed a sensitivity analysis that quantifies the effect of the number of representative days on the trade-off between runtime and accuracy. Because these frameworks are model-agnostic and compatible with any clustering techniques, they can be readily applied to a wide range of energy storage use cases, including utility-scale energy arbitrage, frequency regulation, distribution-level support and residential or commercial systems planning, whenever high-resolution, multi-year analysis is required. Serving as an efficient pre-processing layer, the proposed methods yield quick, reliable insights into battery degradation patterns, facilitating fast prototyping or use in planning optimisation formulation.

Future research could focus on clustering algorithms to capture more diverse load and renewable generation patterns, while also incorporating uncertainties in demand and resource forecasts. Additionally, the results from this degradation analysis serve as a foundation for the development of an accurate battery planning framework, where battery sizing and cost optimisation are performed to identify the most economically and technically viable configurations. Further extensions of this framework may explore its application to alternative battery chemistries and hybrid energy storage systems, ensuring that the representative-days-based approach remains a robust and computationally efficient tool for modern energy storage planning and operational optimisation.

Table A.4
Sensitivity analysis of Proposed-II approaches with increased number of representative days.

Year	Approach	Annual degradation	Runtime	EoL (SoH=0.80)	Difference (+/- than Base)
2014	Base Case	1.878%	109.62 min	10.65 years	N/A
	Proposed – II (5r)	2.419%	0.03 min	8.26 years	(+) 28.8%
	Proposed – II (10r)	2.318%	0.07 min	8.63 years	(+) 23.4%
	Proposed – II (20r)	2.369%	0.09 min	8.44 years	(+) 26.1%
2015	Base Case	1.890%	98.75 min	10.59 years	N/A
	Proposed – II (5r)	2.313%	0.04 min	8.65 years	(+) 22.4%
	Proposed – II (10r)	2.267%	0.07 min	8.82 years	(+) 19.95%
	Proposed – II (20r)	2.109%	0.07 min	9.48 years	(+) 11.6%
2016	Base Case	1.962%	94.24 min	10.19 years	N/A
	Proposed – II (5r)	2.434%	0.03 min	8.21 years	(+) 24.1%
	Proposed – II (10r)	2.544%	0.04 min	7.86 years	(+) 29.7%
	Proposed – II (20r)	2.423%	0.07 min	8.25 years	(+) 23.5%
2017	Base Case	1.905%	111.03 min	10.50 years	N/A
	Proposed – II (5r)	2.350%	0.03 min	8.51 years	(+) 23.4%
	Proposed – II (10r)	2.525%	0.04 min	7.92 years	(+) 32.5%
	Proposed – II (20r)	2.419%	0.09 min	8.26 years	(+) 27.0%
2018	Base Case	1.968%	74.40 min	10.16 years	N/A
	Proposed – II (5r)	2.445%	0.03 min	8.18 years	(+) 24.2%
	Proposed – II (10r)	2.336%	0.06 min	8.56 years	(+) 18.7%
	Proposed – II (20r)	2.362%	0.08 min	8.46 years	(+) 20.0%
2019	Base Case	1.904%	88.77 min	10.51 years	N/A
	Proposed – II (5r)	2.435%	0.03 min	8.21 years	(+) 27.9%
	Proposed – II (10r)	2.325%	0.06 min	8.60 years	(+) 22.1%
	Proposed – II (20r)	2.266%	0.08 min	8.83 years	(+) 19.0%
2020	Base Case	2.094%	59.31 min	9.57 years	N/A
	Proposed – II (5r)	2.320%	0.04 min	8.62 years	(+) 10.8%
	Proposed – II (10r)	2.533%	0.04 min	7.89 years	(+) 21.0%
	Proposed – II (20r)	2.529%	0.07 min	7.91 years	(+) 20.8%
2021	Base Case	1.990%	67.69 min	10.05 years	N/A
	Proposed – II (5r)	2.367%	0.03 min	8.45 years	(+) 18.9%
	Proposed – II (10r)	2.509%	0.06 min	7.97 years	(+) 26.1%
	Proposed – II (20r)	2.283%	0.08 min	8.76 years	(+) 14.7%
2022	Base Case	1.957%	100.92 min	10.22 years	N/A
	Proposed – II (5r)	2.322%	0.04 min	8.61 years	(+) 18.7%
	Proposed – II (10r)	2.085%	0.07 min	9.59 years	(+) 6.5%
	Proposed – II (20r)	2.238%	0.07 min	8.93 years	(+) 14.4%
2023	Base Case	1.957%	99.81 min	10.22 years	N/A
	Proposed – II (5r)	2.333%	0.05 min	8.57 years	(+) 19.2%
	Proposed – II (10r)	2.451%	0.06 min	8.16 years	(+) 25.2%
	Proposed – II (20r)	2.440%	0.07 min	8.19 years	(+) 24.7%

CRediT authorship contribution statement

Shah Mohammad Mominul Islam: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hossein Ranjbar:** Writing – review & editing, Supervision, Investigation, Data curation, Conceptualization. **S. Ali Pourmousavi:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization. **Wen L. Soong:** Writing – review & editing, Supervision.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to improve readability and language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Tables of detailed results

See Tables A.1–A.4.

Data availability

Data will be made available on request.

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